

# Cloud-Native Detection of Cyberattacks Against Satellite Ground Stations: An Open, Labelled Testbed and Two Negative Results

Henrique [Surname]

**Abstract**—The space segment of a satellite mission is expensive to reach; the ground segment that commands it is an ordinary cloud workload with operators, credentials and object storage. Published satellite-security research concentrates on the spacecraft and the radio link, and no labelled corpus exists for measuring ground-segment detection. We present an open testbed: a simulated ground station emitting four correlated planes of evidence, five adversary-emulation scenarios with per-event ground truth, and a cloud-native pipeline of 21 stateful rules plus an unsupervised telemetry baseline. Over 20 independent 24-hour trials it detects all five scenarios in every trial at a precision of 0.997, flagging 0.06% of benign events. The learned baseline makes a sub-threshold behavioural drift visible 42 minutes before the first spacecraft limit violation, raising event-level recall for that scenario from 0.19 to 0.97. Two results contradict our design rationale. First, the multivariate model does not outperform a well-calibrated per-channel detector on that drift: areas under the receiver operating characteristic curve are 0.996 and 0.998. Second, a probe that preserves every marginal distribution while destroying the joint structure, by replaying housekeeping frames from half an orbit earlier, is invisible to both detectors, scoring 0.322 and 0.385 and alerting in 1 of 20 trials, because a baseline fitted over a whole orbit marginalises away the phase information the attack violates. Fitting that baseline on less than one full diurnal cycle also raises the telemetry false-positive rate from 0.2% to 77%. All code and data are public.

**Index Terms**—Satellite ground stations, space systems security, intrusion detection, anomaly detection, cloud computing security, adversary emulation, telemetry.

## I. INTRODUCTION

**A** SATELLITE is difficult to attack directly and difficult to repair once attacked. The ground station that commands it is neither. Modern ground segments are built from the same components as any other cloud workload: identity providers, application programming interfaces, message queues, object storage and the operators who hold the credentials. An adversary who reaches the commanding path from there inherits authority over the vehicle without ever touching a radio.

Recent work has made the space segment measurable. Willbold et al. [1] performed the first systematic software-security analysis of real satellite firmware; Pavur et al. demonstrated large-scale interception of operational satellite links [2] and argued for a research agenda built on reproducible artefacts [3]. Policy has followed, with Space Policy Directive-5 naming the ground segment explicitly as part of the protected

system [4], and the community now has a space-specific adversary taxonomy in SPARTA [5].

What is missing is what made progress possible in network intrusion detection: an open corpus with ground truth. Surveys of network intrusion detection datasets list dozens of options [6]; for the satellite ground segment there is, to our knowledge, none. Ground-segment telemetry and audit logs are operationally sensitive and rarely published, and when they are, they are unlabelled. The practical consequence is that ground-segment detection claims cannot be compared, and that a detector's cost in false positives, analyst time and compute is usually not reported.

This paper contributes a testbed and an evaluation rather than a new detector:

- **An open, labelled ground-station corpus generator** (Section IV). A simulated station emits four correlated planes of evidence, and five adversary-emulation scenarios inject themselves into that timeline, labelling every event they create or modify. Scenarios compose: the stolen session established by the credential stage is the one the tampering and exfiltration stages reuse.
- **A cloud-native detection pipeline** (Section III). 21 stateful rules covering identity, commanding integrity, replay, spacecraft limits and exfiltration, each mapped to an ATT&CK technique [7] and emitted in a standard finding format [8], plus an unsupervised telemetry baseline that runs inside a serverless function with no machine-learning runtime.
- **A measured evaluation with negative results** (Sections V–VI). Twenty independent trials, an ablation separating rules from the model, comparisons against the two alternatives a real mission already has, a probe that isolates joint structure from marginals, a baseline-size sweep, and the compute cost. Two findings contradict the rationale we designed the model around, and we report them as such.

We organise the evaluation around five questions. **RQ1:** does the pipeline detect each scenario, and at what precision? **RQ2:** what does the learned baseline add over the spacecraft's own limit checks, the alternative a real mission already operates? **RQ3:** does the multivariate model add anything over a well-calibrated per-channel detector? **RQ4:** how much clean telemetry does the baseline need before it is usable? **RQ5:** what does the pipeline cost to run?

The author is an independent researcher. Code, data and the experiment harness: <https://github.com/rikosoo/Cloud-Native-Detection-of-Cyberattacks-Against-Satellite-Ground-Stations>.

## II. BACKGROUND AND THREAT MODEL

### A. The ground segment as an attack surface

A ground station converts operator intent into authenticated telecommands and spacecraft state into archived data products. Four kinds of record are produced, and a detector needs all four: *telemetry*, the spacecraft housekeeping channels down-linked during a contact (bus voltage and current, battery state of charge, amplifier and battery temperatures, link margin); *telecommands* (CCSDS telecommand frames [9], authenticated in modern missions by the Space Data Link Security protocol [10], which supplies a keyed authentication tag and an anti-replay sequence counter); *cloud audit records* (who signed in, from where, with what authenticator, and which interfaces they called); and *downlink transfers* (payload products moving into, and potentially out of, the mission archive).

The first two describe the vehicle; the last two describe the enterprise. Most published space-security work addresses the first two, and most enterprise security tooling addresses the last two. An intrusion crosses between them, which is the observation the design in Section III is built on.

### B. Adversaries and trust boundaries

We consider three profiles, following [11] and [12]: the *opportunistic* adversary, commodity tooling and no mission knowledge, high volume and low stealth; the *insider*, a badged operator acting outside their role with legitimate credentials from legitimate networks at legitimate times, so that only *what* they do is anomalous and no network- or identity-provenance rule can see them; and the *state-sponsored* adversary, patient and mission-aware, who targets key material and the commanding path, disables logging early, and prefers degradation indistinguishable from a bus fault, because an operator who believes the spacecraft is unhealthy will act on the wrong runbook.

Four boundaries separate the operator from the vehicle: the human-to-cloud boundary (B1, defended by multi-factor authentication and network policy), the cloud control plane (B2, least-privilege authorisation and audit logging), the up-link (B3, the authentication tag and sequence counter) and the spacecraft itself (B4, on-board limit checks). Table I maps the five attack paths we emulate to SPARTA [5] and ATT&CK [7]. Briefly: *credential\_compromise* sprays passwords, signs in without multi-factor authentication from an untrusted netblock outside duty hours, creates an access key, attaches an administrative policy and deletes the audit trail; *command\_tampering* rewrites opcodes and parameters in flight, forges an authentication tag with the wrong key, issues an out-of-role command and floods the uplink; *replay* retransmits recorded frames verbatim hours later, where the tag still verifies because the adversary never held the key [13]; *anomalous\_behavior* commands a degradation in which every channel stays inside its interface-control limits while the relationships between them break; and *exfiltration* enumerates the archive, reads objects in bulk and copies multiple gigabytes cross-account to an unapproved destination.

We assume the event stream reaches the detector: an adversary who can silently suppress telemetry defeats every rule,

TABLE I  
EMULATED ATTACK PATHS AND FRAMEWORK MAPPING. B1–B4 ARE THE TRUST BOUNDARIES DEFINED ABOVE; THE ATT&CK COLUMN GIVES THE PRIMARY TECHNIQUE.

| Scenario              | Bnd.   | SPARTA   | ATT&CK    |
|-----------------------|--------|----------|-----------|
| Credential compromise | B1, B2 | IA-0004  | T1078.004 |
| Command tampering     | B3     | EX-0012  | T1565.002 |
| Replay                | B3     | EX-0013  | T1557     |
| Anomalous behaviour   | B4     | IMP-0002 | T1565.001 |
| Exfiltration          | B2     | EXF-0003 | T1537     |

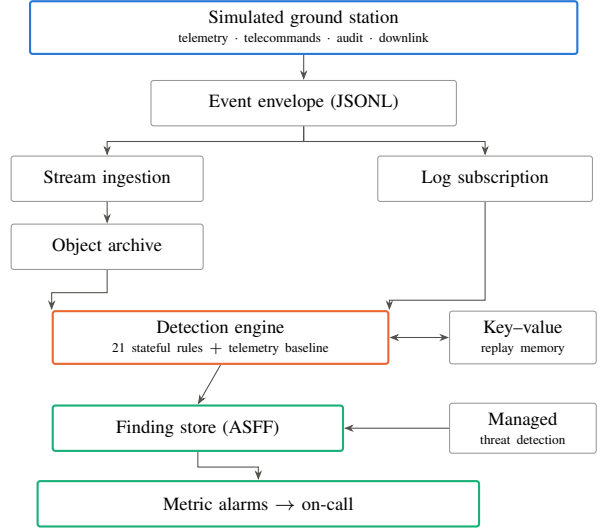


Fig. 1. Pipeline. The detection engine is the same object offline, under test and in the cloud. Managed threat-detection findings are re-scored against the mission asset inventory before entering the same queue.

which is why the deployed pipeline alarms on the *absence* of events. We assume audit logging is on, and treat its disabling as itself a critical finding. We assume telecommand key material is not already stolen; when it is, the integrity rule goes quiet and only the behavioural rules remain, which is why the catalogue does not rest on the authentication tag alone.

## III. SYSTEM DESIGN

Figure 1 shows the pipeline. Three properties drove the design. *One engine, three runtimes*: the detection engine imports no cloud software development kit, and runs unchanged in the offline command-line tool, in the test suite, and inside a serverless function; cloud services are reached only through an event sink interface and a state-store interface. *Offline first*: the full pipeline of generate, fit, detect and score runs with a numerical library and a configuration parser, with no credentials and no network, so a reader without a cloud account can reproduce the artefact. *Measured, not asserted*: every injected event carries a ground-truth label, so precision, recall and detection latency are computed rather than claimed.

### A. Event envelope and rule engine

All four planes share one envelope: an identifier, a type, a coordinated universal time stamp, the station identifier, a plane-specific payload, and (in the lab only) the ground-truth

label. A single stream means a single consumer, and the audit payload deliberately mirrors the field names of a real cloud trail so that the identity rules run unchanged against production logs.

The engine consumes events in timestamp order and emits findings carrying a severity, an ATT&CK technique, the evidence that triggered them and a remediation. Severities and thresholds live in a catalogue separate from the logic, so an operator can retune without a code deployment. State is bounded by time rather than by event count. Three horizons appear: failed-login, command-rate and anomaly-streak state over seconds to minutes; last-login geography per operator over an hour; and accepted sequence counters and authentication tags over the full anti-replay window of 24 hours.

Only the last horizon must be globally consistent. A replayed frame was accepted hours earlier, quite possibly by a different function invocation on a different shard, so replay detection is the one rule that cannot be answered from a single batch. The engine therefore addresses a *replay memory* interface, backed by a dictionary offline and in the cloud by a key–value table with a time-to-live and a conditional write, which makes the deduplication race-free without a lock.

### B. Telemetry baseline

Let  $x \in \mathbb{R}^d$ ,  $d = 7$ , be the vector of housekeeping channels. We fit a single robust Gaussian on telemetry from a run known to be clean. Location and scale are estimated per channel by the median and the normalised median absolute deviation [14], so that the few outliers any real archive contains do not drag the baseline:

$$\hat{\mu}_j = \text{med}(x_j), \quad \hat{\sigma}_j = 1.4826 \cdot \text{med}(|x_j - \hat{\mu}_j|). \quad (1)$$

Channels are standardised,  $z = (x - \hat{\mu}) \oslash \hat{\sigma}$ , and the sample covariance  $S$  of  $z$  is shrunk towards its diagonal in the spirit of [15] with a fixed intensity  $\lambda = 0.15$ , so that the inverse remains well conditioned on short baselines:  $\tilde{S} = (1 - \lambda)S + \lambda \text{diag}(S)$ . The anomaly score of a sample is its squared Mahalanobis distance [16],

$$s(x) = z^\top \tilde{S}^{-1} z, \quad (2)$$

and the detection threshold  $\tau$  is the empirical  $q$ -quantile of  $s$  over the training run, with  $q = 0.999$  throughout. A finding is raised only after  $k = 3$  consecutive samples exceed  $\tau$ ; Section VI shows that this streak requirement is what converts a non-zero per-sample false-positive rate into zero false alerts.

Three properties matter more than accuracy here. The model fits in milliseconds, serialises to a 1.5 KB object that a serverless function loads without a machine-learning layer, and decomposes: the per-channel contributions to (2) are available for free, so a finding can name the channels that moved. An analyst at 03:00 needs that more than a fractionally better score, and Section VI shows the fractionally better score was not on offer anyway.

## IV. IMPLEMENTATION

The simulated spacecraft is a 95-minute low-Earth orbit with a 35% eclipse fraction and an 11-minute contact window

per orbit. The bus model is small but deliberately *coupled*: illumination drives the power budget, the power budget and eclipse state drive the thermal channels, and transmit duty and slew rate drive the amplifier temperature and the link margin. Coupling is what makes RQ3 meaningful, because an adversary who moves one channel while leaving the others untouched breaks the relationships long before breaking any single limit. The station drives that model at a 30-second cadence and adds the three ground-segment planes: shift handovers produce console sign-ins (service identities federate through role assumption instead), contact windows produce telecommands authenticated with a keyed tag over a canonical serialisation plus a monotonic sequence counter, and each pass ends with a payload downlink.

Attacks are applied *after* the nominal run, so “normal” is defined in one place. Each injector receives the timeline and a context object, and returns the timeline with its events inserted or mutated and labelled. Injectors therefore compose. The credential stage registers the hijacked session that the tampering and exfiltration stages reuse, which produces genuine cross-plane correlation rather than five independent incidents. Because injection is a transformation of a clean timeline, the same seed also yields the same corpus byte for byte.

The deployed pipeline maps each stage to a managed service: stream ingestion partitioned by station, continuous archival to compressed object storage, the engine in a serverless function with partial-batch failure reporting, the replay memory in a key–value table with conditional writes and a time-to-live, the identity plane from a cloud audit trail carrying both management and object-level events, a managed threat-detection service re-scored by a second function against the mission asset inventory, and findings in a single store in a standard format [8].

Three defects appeared only once that cloud path was exercised against emulated services. Each is a silent failure that a simulation-only evaluation would not surface. The log-ingestion interface rejects events older than 14 days *inside a successful response*, so the reproducible corpus, whose timestamps are fixed for determinism, vanished into an empty log group with a zero exit code. Declaring advanced event selectors on the audit trail *replaces* the default management-event selector, so a data-events-only configuration would have recorded no sign-ins or credential creations at all, silently disabling the entire identity rule family. And the stream-to-function mapping advertises partial-batch failure reporting, but a handler that does not return the failed record identifiers makes one malformed record fail and retry an entire 200-record batch. All three are fixed and covered by regression tests.

## V. EXPERIMENTAL METHODOLOGY

### A. Corpus and protocol

Table II describes one 24-hour run. Telemetry dominates by design, at one sample every 30 seconds; the identity and downlink planes are sparse, which is exactly the regime in which a per-event false-positive rate matters. The label distribution is uneven and we do not smooth it: replay and

TABLE II

ONE 24-HOUR CORPUS: EVENTS BY PLANE, AND EVENTS CARRYING EACH ATTACK LABEL. LABELLED EVENTS OVERLAP THE PLANES THEY OCCUR IN.

|                       | Events      | %            |
|-----------------------|-------------|--------------|
| <i>Evidence plane</i> |             |              |
| telemetry             | 2880        | 80.4         |
| telecommand           | 583         | 16.3         |
| audit                 | 102         | 2.8          |
| downlink              | 19          | 0.5          |
| <i>Attack label</i>   |             |              |
| credential_compromise | 11          | 0.3          |
| command_tampering     | 43          | 1.2          |
| replay                | 8           | 0.2          |
| anomalous_behavior    | 404         | 11.3         |
| exfiltration          | 7           | 0.2          |
| Benign                | 3111        | 86.8         |
| <b>Total</b>          | <b>3584</b> | <b>100.0</b> |

exfiltration label single-digit numbers of events, while `anomalous_behavior` labels several hundred telemetry samples because the drift it injects persists for hours. Aggregate recall would therefore be dominated by one scenario, so we report recall per scenario throughout and never average it.

We run 20 independent trials. Trial  $i$  fits the telemetry baseline on a clean 24-hour run generated with seed  $i + 10^4$ , then detects on an attacked run generated with seed  $i$ . Training and evaluation data are therefore disjoint and independently generated, and no attacked sample ever reaches the model. All trials use the same rule catalogue and the same  $q = 0.999$  threshold quantile; nothing is tuned per trial or per scenario. All experiments run single-threaded, and the artefact's result record stores the library versions and platform string alongside every number reported here.

### B. Metrics, ablation and reference detectors

A finding is a *true positive* when the event it names carries an attack label, and a *false positive* otherwise. We report the *detection rate*, the fraction of trials in which a scenario produced at least one true positive, which is the question an operator asks; *event-level recall*, the fraction of a scenario's labelled events that produced at least one finding, which is the stricter question and the one where a persistent scenario can be "detected" while most of its events go unflagged; *precision* and the *false-positive rate*, the latter over benign events rather than over findings, so that it is comparable across scenarios of different sizes; and *mean time to detect* (MTTD), the interval between a scenario's first labelled event and the first finding attributable to it.

We evaluate three configurations: *rules only* (the catalogue with the model-driven rule removed), *model only* (that rule alone), and *rules + model*. For RQ2 and RQ3 the model is compared against the two alternatives a real mission already has. The first is the *interface-control limits*: the spacecraft's own red lines, encoded as a rule. This is the deployed status quo, and the only honest comparison for whether a learned baseline earns its place. The second is the *best single channel*: a per-channel detector that already knows the robust centre

TABLE III

EFFECTIVENESS OVER 20 INDEPENDENT TRIALS. DETECTION RATE IS THE PERCENTAGE OF TRIALS IN WHICH THE SCENARIO PRODUCED AT LEAST ONE FINDING; RECALL IS PER LABELLED EVENT, FOR THE FULL PIPELINE (*both*) AND FOR EACH ABLATED HALF. STANDARD DEVIATIONS ARE BELOW 0.01 FOR EVERY SCENARIO.

| Scenario                                     | Det.<br>% | Recall |       |       | MTTD<br>(s) |
|----------------------------------------------|-----------|--------|-------|-------|-------------|
|                                              |           | both   | rules | model |             |
| Credential compromise                        | 100       | 0.82   | 0.82  | 0.00  | 18          |
| Command tampering                            | 100       | 1.00   | 1.00  | 0.00  | 0           |
| Replay                                       | 100       | 1.00   | 1.00  | 0.00  | 0           |
| Anomalous behaviour                          | 100       | 0.97   | 0.19  | 0.97  | 342         |
| Exfiltration                                 | 100       | 1.00   | 1.00  | 0.00  | 0           |
| Precision (rules + model): $0.997 \pm 0.002$ |           |        |       |       |             |
| False positives: 0.06% of benign events      |           |        |       |       |             |

and scale of every channel and takes the largest absolute standardised deviation,  $u(x) = \max_j |z_j|$ , thresholded at the same training quantile. It is intentionally the strongest univariate strawman, since it is given everything the multivariate model has except the covariance.

### C. The phase-swap probe

The drift injected by `anomalous_behavior` pushes several channels well outside their own robust scale, which means it is detectable marginally and cannot separate "multivariate" from "well-calibrated univariate". To test the design claim directly we construct a probe that preserves every marginal exactly while destroying the joint structure. In a clean run, each telemetry vector in a 50-minute window is replaced by the vector observed exactly half an orbit earlier, channel for channel. Every individual value is one the spacecraft genuinely produced and lies inside its own learned marginal distribution. The *combination*, eclipse-era power draw beside sunlit-era temperatures, is physically impossible. Orbit phase is not among the model's features, so no per-channel test can see the substitution at all, and only a model of the relationships between channels can.

The probe also models a real technique: replaying recorded housekeeping frames to mask a fault or the effect of a malicious command. We keep it as a controlled probe rather than a sixth deployed scenario so that the headline effectiveness numbers describe the shipped system.

## VI. RESULTS

### A. RQ1: Effectiveness

Table III and Fig. 2 give the main result. All five scenarios are detected in all 20 trials, at a precision of  $0.997 \pm 0.002$  and a false-positive rate of 0.06% of benign events. The four scenarios that leave discrete evidence, such as a forged tag, a reused counter, a policy attachment or a transfer to an unapproved destination, reach event-level recall of 1.00 with zero variance and an MTTD of 0 seconds, because the first event of the scenario is itself the violation.

The ablation shows the two halves are complementary and non-overlapping. The model alone detects nothing outside the telemetry plane, having nothing to say about a forged

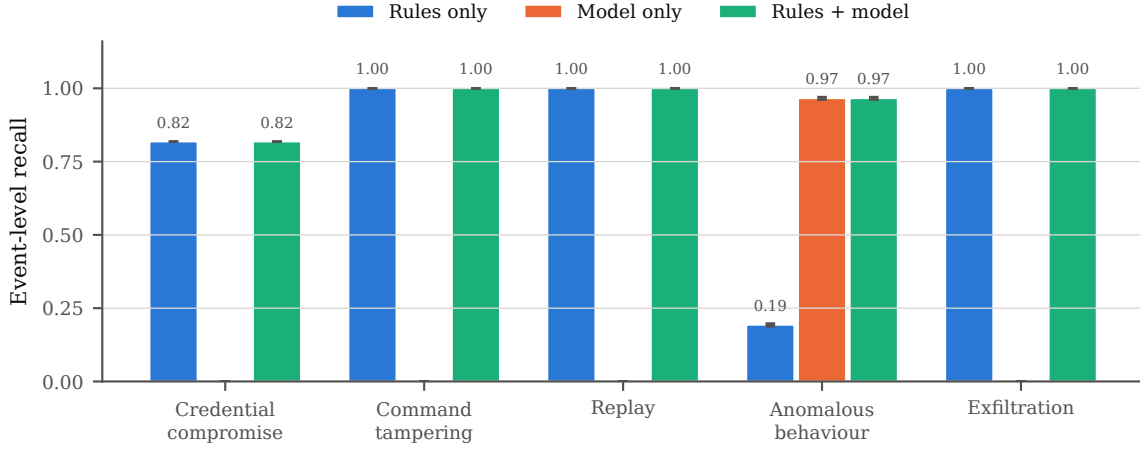


Fig. 2. Event-level recall by scenario and configuration, mean over 20 trials with standard-deviation bars. Absent bars are zero. The rule catalogue and the learned baseline are complementary rather than overlapping: neither configuration alone covers the corpus.

authentication tag, and the rules alone reach only 0.19 recall on the behavioural drift. Together they cover the corpus. The model configuration attains a precision of 1.000 with zero false positives in any trial, which is the effect of the streak requirement of Section III-B. The per-sample false-positive rate at the operating point is 0.2%, but isolated exceedances never survive three consecutive samples.

The residual false positives are concentrated in two rules, which we report individually rather than in aggregate. The impossible-travel rule fires 1.00 times per trial, always on the *legitimate* operator sign-in that follows the attacker's. That is correct behaviour which a label-based metric is obliged to count as an error. The command-rate rule fires 1.58 times in the trials where it fires at all, on nominal bursts that cross a per-operator ceiling set too tightly. The first is a metric artefact; the second is a genuine tuning defect.

#### B. RQ2: What the baseline adds over spacecraft limits

This is the comparison that matters operationally, because the limit check is what a mission already runs. Across trials, the learned baseline raises a finding  $42 \pm 2$  minutes before the first limit violation (range 37–44 minutes, and in no trial did the limit fire first). Event-level recall for the drift rises from 0.19 to 0.97, and MTTD falls from 47 to 6 minutes.

The interpretation is narrow. A limit check tells an operator the spacecraft is *already* outside its design envelope, whereas the baseline tells them it is heading there. Forty minutes is the difference between diagnosing a degradation and reacting to an alarm. This is not evidence that the model is a good anomaly detector in general, only that it is a better early indicator than a red line.

#### C. RQ3: Multivariate versus univariate detection

*On the injected drift, the covariance buys nothing.* On the telemetry plane of the attacked runs, over 20 trials, the multivariate detector achieves an area under the receiver operating characteristic curve of  $0.996 \pm 0.001$  and the best single channel  $0.998 \pm 0.000$ . The univariate detector is marginally *better*,

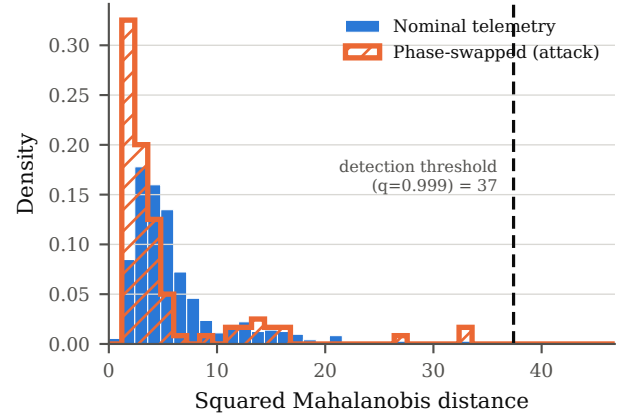


Fig. 3. Score distributions under the phase-swap probe. Replayed housekeeping frames from half an orbit earlier sit inside, and in fact below, the nominal distribution, because orbit phase is marginalised away by a baseline fitted over a whole orbit. Texture is redundant with colour for grayscale printing.

the gap is smaller than the between-trial spread, and over the full range the two curves are visually indistinguishable. Our design rationale, that correlation is what makes a sub-threshold attack visible, is not supported for this attack.

The explanation is in the attack, not the model. The injected drift moves the slow rate, bus current and amplifier temperature several robust standard deviations from their own centres, so each channel is individually anomalous and a per-channel test suffices. A fair reading is that our behavioural scenario is easier than intended, and that the experiment does not test the claim.

*On an attack that isolates the joint structure, both detectors fail.* The phase-swap probe of Section V-C does test it, and the result is worse than inconclusive. Fig. 3 shows why: the swapped vectors do not simply fail to exceed the threshold; they score *lower* than nominal telemetry, at a median squared distance of 2.9 against 4.6 for nominal samples. Both detectors are below chance ( $0.322 \pm 0.026$  multivariate,  $0.385 \pm 0.019$  univariate) and the attack is flagged in only 1 of 20 trials.

This is a structural limitation, not a tuning problem, and

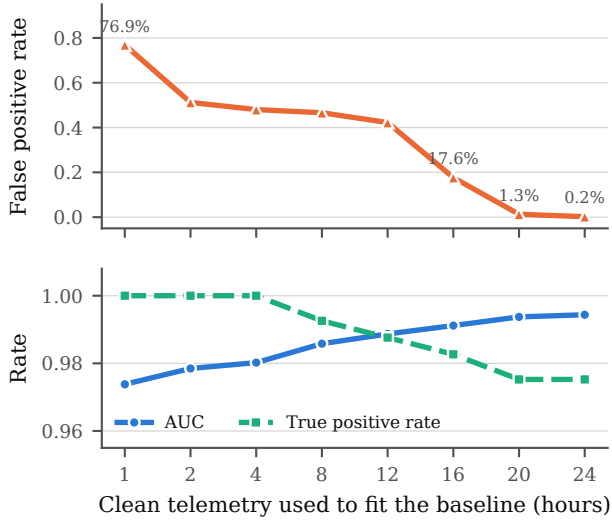


Fig. 4. Baseline-size sensitivity. Upper panel: telemetry false-positive rate at the deployed operating point. Lower panel: area under the curve and true-positive rate. The false-positive rate collapses only when the training window covers a full diurnal cycle; discrimination barely changes.

no threshold fixes it. A single Gaussian fitted over an entire orbit learns the *average* relationship between channels, marginalising over the phase; any vector that occurred at any phase is therefore in-distribution. That the swapped samples score below the median follows: mid-orbit values are denser than the extremes the true samples occupy. An adversary who replays valid old telemetry hides inside the learned baseline, and makes the spacecraft look healthier than it is. Because the failure is below chance it is also *exploitable*: the same replay drives the anomaly score down while a fault progresses.

The fix is to condition on phase rather than to marginalise it: model the residual from a phase-dependent expectation, or fit a mixture indexed by orbit phase, including phase as an explicit conditioning variable. We report the failure rather than implementing the fix here, because the probe is the contribution. It is a cheap, reusable test that any spacecraft telemetry detector can be subjected to, and we would expect several published detectors to fail it. The implication appears to generalise: any detector that fits a phase-marginal distribution over a periodic system inherits the same blind spot, including detectors considerably more sophisticated than ours [17], [18], unless they condition on phase explicitly.

#### D. RQ4: How much clean telemetry the baseline needs

Fig. 4 sweeps the size of the training window. The finding is a sharp threshold rather than a gradual improvement: fitting on one hour yields a telemetry false-positive rate of 77%, reaching 0.2% only once the window covers a full 24 hours. Discrimination improves marginally across the sweep (area under the curve 0.974 to 0.994) and the true-positive rate declines slightly as the threshold rises, so neither explains the collapse.

The cause is coverage, not sample size. A window shorter than the orbital and diurnal cycle observes only part of the state space, covering some eclipse fractions, contact geometries and

thermal regimes but not others, so the quantile threshold is calibrated against a distribution the deployed model will not see. The operational rule this implies is concrete: a baseline must span at least one full cycle of every periodicity the mission has, and a detector that reported only the area under the curve would have concluded, wrongly, that one hour of data is nearly as good as a day.

#### E. RQ5: Cost and alert volume

The engine processes 78 880 events per second with the model enabled ( $12.7 \mu\text{s}$  per event) and 189 012 without it; fitting the baseline on a day of telemetry takes 0.02 s and produces a 1.5 KB model. At a 30-second telemetry cadence, one core is approximately five orders of magnitude faster than one station requires. The constraint is transport and storage, not detection.

The analyst-facing load is a different matter. The mean of 613 findings per simulated day is a misleading summary: the load is extremely bursty, with nothing for twenty hours and a peak of roughly 176 findings in a single hour, of which 26 per day are critical. Aggregate false-positive rates say little about whether a queue is workable, and the base-rate fallacy [19] applies with full force at these volumes. The practical mitigation in our design is severity discipline plus correlation: critical findings page, and the chains in Table I mean an operator sees one incident rather than four unrelated alerts.

## VII. DISCUSSION AND LIMITATIONS

### A. Rules and models do different jobs

Deterministic rules catch the four highest-impact scenarios, namely forged commands, replay, privilege escalation and exfiltration, immediately, with no variance and no training data. They do so because the mission has an explicit policy to state, such as that a tag must verify, that a counter must not repeat, or that a role may not issue a given opcode, and a rule is the natural expression of a policy. Learned detection earns its place only where no policy exists to state, on behaviour that is legal at every instant and wrong in aggregate. Our two negative results point the same way: the marginals carry most of the signal a static baseline can extract, and the joint structure it learns is too coarse to catch an adversary who respects them.

The three silent failures in Section IV have a common shape. The cloud interface accepted the request, returned success, and discarded the data or the configuration intent. Such failures are invisible in a detection pipeline, because an empty log group and a quiet queue look identical to a quiet day.

### B. Threats to validity

*The corpus is synthetic.* This is the dominant threat to validity. Our spacecraft model is a sine-wave illumination profile with first-order thermal lags; real housekeeping telemetry has instrument nonlinearity, sensor faults, seasonal beta-angle drift and operator idiosyncrasy that we do not reproduce, so absolute numbers should be read as properties of this corpus rather than of a mission. The comparative and structural

findings (the rules/model complementarity, the ordering of lead times, the phase-swap blindness and the coverage threshold) depend on the *structure* of the data rather than its fidelity, and we expect them to transfer. The precision figure does not.

*Attacks are injected, not executed*, and the adversary is non-adaptive: it does not observe the detector and adjust, which is the weakest assumption in any static evaluation of this kind [20]. The phase-swap probe is our one concession to an adaptive adversary, and it succeeded. Recall is also per event, so a scenario labelling 404 events and one labelling 7 are not comparable, and the impossible-travel false positive in Section VI is correct detector behaviour scored as an error; we report per-scenario numbers to keep this visible rather than resolving it.

*Scope*. One station, one spacecraft, one region; cross-station correlation is a stronger signal than either station alone and we do not evaluate it. The cloud evaluation is emulated in process, so it cannot surface authorisation-policy mistakes, service quotas or eventual consistency, and the infrastructure code has not been applied against a live account, so the deployed pipeline's cost and latency are unmeasured. We also evaluate one baseline family; isolation forests [21], sequence models [17] and probabilistic clustering [18] are the obvious comparisons and are left to future work, and the model sits behind a score-and-threshold interface so they can be substituted and scored on the same labels.

### VIII. RELATED WORK

*Space systems security*. Pavur and Martinovic surveyed the field and argued that its central problem is the absence of reproducible artefacts and shared testbeds [3]; this paper responds to that argument for one part of the system. Pavur et al. demonstrated practical interception of operational maritime satellite links [2], and Willbold et al. analysed real satellite firmware and found systematic absence of authentication on the commanding path [1]. Falco [12] and Manulis et al. [11] set out security principles for space systems and for the commercial segment; SPARTA [5] and ATT&CK [7] provide the taxonomies we map to, and the commanding model follows the CCSDS telecommand protocol [9] and its security layer [10]. Our focus differs in being the ground segment's cloud footprint, and in being evaluated against labels rather than argued.

*Telemetry anomaly detection and evaluation methodology*. Detecting faults in housekeeping telemetry is mature: Hundman et al. applied recurrent networks with nonparametric dynamic thresholding to spacecraft channels [17], Yairi et al. combined probabilistic clustering with dimensionality reduction [18], and Chandola et al. survey the wider area [22]. That literature targets *faults*, and its evaluations assume anomalies are not adversarially chosen; our phase-swap result suggests the assumption is load-bearing, because a fault will not respect the marginals and an adversary will. On the reporting side, Sommer and Paxson's critique of machine learning for intrusion detection [20] and Axelsson's base-rate analysis [19] shape how we present results: per scenario rather than aggregate, false positives over benign events, and alert

volume as a first-class result. Ring et al. survey the network intrusion detection datasets [6] whose absence here motivated the corpus, and automated adversary emulation in the style of CALDERA [23] informed the composing-injector design.

### IX. CONCLUSION

We built an open, labelled testbed for detecting attacks on satellite ground stations, and a cloud-native pipeline that detects five emulated scenarios in every one of 20 trials at  $0.997 \pm 0.002$  precision. Four further results matter more than that number. Deterministic policy rules, not learned detection, carry the high-impact scenarios. The learned baseline earns its place on exactly one of the five scenarios, where it buys 42 minutes over the spacecraft's own limits. Its multivariate machinery does not outperform a well-calibrated per-channel detector on that scenario. And both detectors score below chance against an adversary who replays valid telemetry from a different orbit phase; that last finding is structural, applies to any phase-marginal baseline, and is testable in a few lines of code. The artefact is public, deterministic, and runs offline in seconds, so each of these claims can be checked and, we hope, improved on.

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**Ethical considerations.** No real ground station, spacecraft or third-party system was touched. All telemetry, credentials and audit records are synthetic; the cryptographic keys in the repository are lab constants, labelled as such, and unsuitable for operational use. The attack scenarios describe adversary behaviour at the level of detection engineering and contain no exploit code and no vulnerability in any real product.

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